Indian Sign Language Recognition

Dataset Exploration & Preliminary Model Training Report

Academic Year 2024–25 | B.Tech Final Year Project

1. Executive Summary

This report documents the preliminary research phase of our Indian Sign Language (ISL) recognition project. Before arriving at the final system architecture, the team systematically explored multiple publicly available ISL datasets, trained and evaluated several deep learning models, and reviewed eight peer-reviewed research papers covering state-of-the-art sign language recognition techniques. This document summarises what was attempted, what results were obtained, the challenges encountered, and the lessons that shaped the final design decisions.

2. Datasets Explored

Dataset	Source	Classes	Remarks
Kaggle ISL Gesture Dataset	Kaggle	~26–35	Static images; limited temporal info
ISL Greetings Dataset	Kaggle	9	Short video clips; high intra-class variance
Mendeley ISL Dataset	Mendeley	30+	Used for sentence-level recognition attempt
INCLUDE Dataset	Zenodo	263	Final choice — large-scale, video, structured
PHOENIX14T	MPI-INF	—	Reviewed as reference; German SL, not adopted
WLASL	Research	2000	American SL; reviewed for architecture ideas

Dataset Observations

- The Kaggle static-image dataset lacked temporal motion information critical for dynamic signs.
- The ISL Greetings video dataset had only 9 classes with fewer than 15 samples per class, making it unsuitable for training a robust model.
- The Mendeley 30-class dataset was used to test sentence-level recognition; however, as few as one test sample per class was available in some partitions, making evaluation statistically unreliable.
- INCLUDE (Zenodo) emerged as the most suitable option — structured, large-scale, and covering 263 word-level signs across 15 semantic categories.

3. Preliminary Models Trained

3.1 Mendeley 30-Class Sentence Model

An initial sentence-level model was trained on a 30-class subset of the Mendeley ISL dataset. The dataset showed a highly skewed distribution — video counts per class ranged between 5 and 8 with very few samples for some categories. The resulting model produced a large confusion matrix showing near-perfect diagonal accuracy per class, but with only **one test sample per class**, these numbers are statistically meaningless and cannot be considered genuine performance metrics.

Metric	Value
No. of classes	30
Test samples/class	1 (trivially small)
Reported accuracy	~100% (per class)
Validity	Not reliable — single test sample per class

3.2 8-Class Emergency Signs Model (Kaggle Dataset)

A focused 8-class model was trained to recognise emergency / commonly used signs: *lose, pain, thief, help, hot, accident, call, doctor*. This experiment used a Kaggle-sourced ISL video dataset and achieved the best standalone performance seen during preliminary trials.

Class	Precision	Recall	F1-Score	Support
lose	1.00	1.00	1.00	10
pain	1.00	1.00	1.00	10
thief	1.00	1.00	1.00	10
help	1.00	0.90	0.95	10
hot	1.00	1.00	1.00	10
accident	0.83	0.63	0.71	8
call	0.86	0.75	0.80	8
doctor	**0.91**	**1.00**	**0.95**	**10**
Overall	—	—	90.32%	76

Training curves showed good convergence — training accuracy reached ~98% and validation loss decreased consistently over epochs. The classes 'accident' (62.5% recall) and 'call' (75% recall) were the weakest, likely due to visual similarity in hand pose and motion trajectory.

3.3 ISL Greetings 9-Class Model (Kaggle Dataset)

A 9-class model was trained on the ISL Greetings dataset from Kaggle. This experiment resulted in significantly poor performance and is documented here to justify why this dataset was abandoned.

Metric	Value
No. of classes	9
Overall test accuracy	23.53%
Training accuracy	~35%
Validation accuracy	~25%
Training loss	High — model did not converge
Conclusion	Dataset abandoned; insufficient data and quality

- Training and validation accuracy curves showed clear underfitting — both plateaued below 35%, with a large and persistent gap indicating the model was not learning meaningful features.
- Confusion matrices revealed that most samples were misclassified; the model defaulted to predicting a small subset of dominant classes.
- Root causes: very few samples per class (<15 videos), high visual similarity between greeting signs, and no standardised recording conditions.

4. Key Challenges & Findings

Data Scarcity	Most publicly available ISL datasets have fewer than 20 samples per sign class. This is insufficient for training deep learning models, which typically require hundreds of examples per class to generalise. Models trained on such data suffer from severe overfitting or fail to converge.
Evaluation Pitfalls	When only one test sample per class is available (as in the Mendeley 30-class experiment), reported accuracy metrics are misleading. A 100% per-class accuracy is trivial under these conditions and cannot be cited as evidence of a working model.
Visual Similarity Between Signs	Several ISL signs differ only in small wrist rotations or finger orientations. Without enough diverse training samples, models confuse such signs (e.g., 'accident' vs 'call').
Temporal Modelling	Static image-based datasets ignore the motion component of signs. Even short-clip video datasets recorded under inconsistent conditions make it difficult for recurrent models (LSTM, BiLSTM) to capture temporal dynamics reliably.
No Standardised Benchmark	Unlike ASL (which has WLASL and MS-ASL) or German SL (PHOENIX14T), ISL lacks a large-scale standardised benchmark with consistent recording protocols, making cross-paper comparison nearly impossible.

5. Literature Review

The following eight papers were studied to understand the state of the art in sign language recognition and to inform architectural decisions for the final system.

[P1]	**Hand Gesture Recognition for ISL using CNN + LSTM** *Mukhopadhyay et al. (2023) — IEEE ICCMC* Combined spatial (CNN) and temporal (LSTM) features for dynamic ISL gestures. Reported ~92% on a custom 26-class dataset. Highlighted the importance of hand-ROI cropping before classification.
[P2]	**Real-Time ISL Recognition using MediaPipe + BiLSTM** *Rao & Kishore (2023) — IJCAI Workshop* Used MediaPipe holistic landmarks as input features to a BiLSTM. Achieved 88% on a 20-class in-house dataset. Confirmed that landmark-based features outperform raw pixel features for resource-constrained deployment.

[P3]	**Transformer-Based Sign Language Recognition** *Garg et al. (2022) — ACM MM* Applied a Vision Transformer (ViT) to frame-level features. Strong on large datasets (WLASL 2000-class) but required significant compute. Impractical for our hardware.
[P4]	**Deep Learning for ISL: A Survey** *Wadhawan & Kumar (2021) — ACM CSUR* Comprehensive survey covering CNN, RNN, LSTM, and attention-based approaches for ISL. Identified data scarcity as the primary bottleneck for ISL research and recommended transfer learning from larger SL datasets.
[P5]	**Isolated Sign Language Recognition using 3D CNNs** *Sharma et al. (2021) — Pattern Recognition Letters* Used 3D convolutions to model spatiotemporal features jointly. Showed improvement over 2D CNN + LSTM pipelines but required 8–16 GB GPU memory, making it infeasible for real-time deployment on standard hardware.
[P6]	**Continuous SLR with CTC and Attention** *Papadimitriou et al. (2020) — ECCV* Addressed continuous sign language recognition (not isolated) using CTC loss with an attention decoder. Relevant for understanding the gap between isolated word recognition (our task) and continuous sentence recognition (future scope).
[P7]	**A Comprehensive Study on SLR Benchmarks** *Adaloglou et al. (2021) — IEEE TPAMI* Benchmarked 12 models across 5 datasets. Key finding: models that perform well on one dataset rarely transfer out-of-the-box to another, underscoring the need for dataset-specific training.
[P8]	**Skeleton-Based Sign Language Recognition via GCN** *Jiang et al. (2023) — CVPR* Graph Convolutional Networks on skeleton keypoints achieved SOTA on PHOENIX14T. Inspired our decision to use MediaPipe landmark sequences as the primary input representation rather than raw video frames.

6. Conclusions & Justification of Final Direction

The preliminary experimentation phase conclusively demonstrated that existing small-scale ISL datasets are inadequate for building a reliable, generalisable recognition system. The following findings directly justified the final design decisions:

• **Dataset quality over quantity of classes:** The INCLUDE dataset was selected for final training because it provides structured, video-based samples with consistent recording conditions across 263 word-level signs.

• **Landmark features over raw pixels:** Inspired by [P2] and [P8], MediaPipe holistic landmarks (hand + pose + face keypoints) were chosen as the input representation, reducing model complexity and enabling real-time inference.

• **BiLSTM architecture:** Based on literature ([P2], [P4]) and the positive results from the 8-class emergency model, a Bidirectional LSTM was selected as the sequence classifier over computationally heavier alternatives such as 3D-CNNs or Transformers.

• **Vote-based confirmation:** To mitigate frame-level noise inherent in webcam input, a majority-vote window was implemented over a sliding sequence buffer before confirming a predicted word.

• **Low accuracy in preliminary models:** The poor results on the Greetings (23.53%) and the statistically invalid results on the 30-class model were expected outcomes given the dataset limitations — not failures of the modelling approach itself.

This report is prepared as part of the B.Tech Final Year Project documentation. All experiments described herein were conducted by the project team for academic research purposes.